Software Requirements Specification (SRS)

Nazoru Mobile Application

Version 1.0 — Based on IEEE 830 Standard

---

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification is to define the functional and non-functional requirements of the **Nazoru** mobile application. This document serves as a reference for developers, designers, project managers, and stakeholders to ensure consistent understanding throughout the development lifecycle.

1.2 Scope

Nazoru is a hybrid mobile application that enables users to trace drawings by displaying a semi-transparent image over the camera feed. Its main objective is to function as a portable *camera lucida* for artists, students, and beginners.

1.3 Definitions

- **AR Lite:** Simple augmented overlay without spatial tracking or 3D anchoring.
- **HUD:** Heads-up display; interface elements layered on top of the screen.
- **Capacitor:** A tool that allows hybrid applications to access native device features.

---

2. Overall Description

2.1 Product Perspective

Nazoru is built using **Ionic Vue** and packaged with **Capacitor** to run on Android devices. It communicates with a **NestJS** backend that uses **Prisma** as the ORM and **MySQL** as the main database. The application integrates the device camera and applies a transparent overlay to assist with tracing.

2.2 User Characteristics

The application is intended for:

- **Artists** who need visual tracing assistance
- **Students** practicing drawing
- **Beginners** learning basic artistic tracing techniques

Users require no technical knowledge to operate the app.

2.3 Constraints

- Requires **camera permission**.
- Only supports **Android 8.0+**.
- Performance depends on the device's camera capabilities.
- Internet connection required for registration and login, but guest mode is available offline.

---

3. Functional Requirements

RF-01: The system must allow user registration.
RF-02: The system must allow users to log in with email and password.
RF-03: The system must support guest access without an account.
RF-04: Users must be able to select an image from the device gallery.
RF-05: The app must activate the camera in full-screen mode.
RF-06: The app must display the selected image as an overlay.
RF-07: Users must be able to adjust image opacity using a slider.
RF-08: Users must be able to lock the image position.
RF-09: The app must offer a mirror (horizontal flip) mode.
RF-10: The app must allow toggling the camera flash.
RF-11: Users must be able to scale and move the image using gestures.
RF-12: The system must allow users to log out.

---

4. Non-Functional Requirements

RNF-01 Usability

The interface must be simple, intuitive, and accessible to inexperienced users.

RNF-02 Performance

- Overlay rendering should maintain at least **30 FPS**.
- App startup time must remain under **3 seconds** on supported devices.

RNF-03 Security

- Passwords must be encrypted using **bcrypt hashing**.
- Communication between frontend and backend must occur over HTTPS.

RNF-04 Compatibility

- The app must adjust to various screen sizes and resolutions.
- Supported OS: **Android 8.0 or higher**.

RNF-05 Availability

- The system should maintain **99% uptime** when backend services are active.
- Core overlay functionality must work offline (guest mode).

---

5. User Stories

- *As a user, I want to register so I can save my information and preferences.*
- *As a guest, I want to access the app instantly without creating an account.*
- *As an artist, I want to adjust opacity so I can trace more accurately.*
- *As a beginner, I want to lock the image to avoid accidental movements.*

---

6. Architecture

- **Frontend:** Ionic Vue (Vue 3 + Capacitor)
- **Backend:** NestJS with Prisma ORM
- **Database:** MySQL
- **Mobile:** Native Android theme with transparent camera view using Capacitor plugins

---

7. Database Structure

User Table

Field	Type	Description
id	INT (PK)	Unique identifier
name	VARCHAR	User’s name
email	VARCHAR	Unique email address
password_hash	VARCHAR	Hashed password using bcrypt

---

8. Application Flow

1. **Home Screen**
2. **Register / Login / Guest Access**
3. **How-It-Works Explanation**
4. **Image Selection**
5. **Camera View with Overlay**
6. **Drawing Assistance Controls (opacity, lock, mirror, flash, scaling)**

---

9. Conclusions

Nazoru fulfills its objective of functioning as a digital *camera lucida*, helping users trace drawings efficiently through an intuitive overlay system. Its hybrid architecture ensures flexibility, accessibility, and smooth performance across compatible Android devices.